

# BIPASHA BANERJEE

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## CONTACT INFORMATION

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**GOOGLE SCHOLAR:** <https://scholar.google.com/citations?user=B8WSVAEAAAAJ&hl=en>

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## EDUCATION

**Ph.D. COMPUTER SCIENCE, VIRGINIA TECH, BLACKSBURG, VA** 2024  
**Dissertation:** Banerjee, B. (2024). Improving Access to ETD Elements Through Chapter Categorization and Summarization, Ph.D. Thesis, Virginia Tech, 2024. <https://hdl.handle.net/10919/120890>  
**M.S. COMPUTER SCIENCE, VIRGINIA TECH, BLACKSBURG, VA** 2022  
**B.Tech. COMPUTER SCIENCE & ENGINEERING**  
**WEST BENGAL UNIVERSITY OF TECHNOLOGY, KOLKATA, INDIA** 2015

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## PROFESSIONAL EXPERIENCE

**ASSISTANT PROFESSOR,** August 2024 - Present  
**AI RESEARCH SCIENTIST, DIGITAL LIBRARIES** University Libraries, Virginia Tech, Blacksburg, VA  
**GRADUATE RESEARCH ASSISTANT** August 2019 – August 2024  
University Libraries, Virginia Tech, Blacksburg, VA  
**GLOBAL PRODUCT AND TECHNOLOGY INTERN** May – August 2019  
ADP, Alpharetta, GA  
**GRADUATE TEACHING ASSISTANT** January – May 2019  
Computer Science, Virginia Tech, Blacksburg, VA  
**ASSISTANT SYSTEMS ENGINEER - FULL TIME** April 2016 – November 2017  
TATA Consultancy Services, Kolkata, India

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## RESEARCH INTERESTS

I am an Assistant Professor at Virginia Tech, where I currently work as an AI Research Scientist, Digital Libraries for the University Libraries. My research mainly focuses on digital libraries, natural language processing, machine learning, artificial intelligence, and scholarly big data.

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## RECENT PUBLICATIONS & TALKS

- Salsabil, L., Obadage, R. R., Banerjee, B., Abeysinghe Y., Alam, S., Färber, M., Ingram, W. A., Fox, E.A. & Wu, J. Context-Based URL Classification for Open Access Datasets and Software in Scholarly Document. In 2025, The 25th ACM/IEEE Join Conference on Digital Libraries (JCDL) DOI: [10.1109/JCDL67857.2025.00031](https://doi.org/10.1109/JCDL67857.2025.00031)
- Ingram, W. A, Banerjee, B. & Fox, E.A. Learning from LLM Disagreement in Retrieval Evaluation. In 2025, The 25th ACM/IEEE Join Conference on Digital Libraries (JCDL). DOI: [10.1109/JCDL67857.2025.00024](https://doi.org/10.1109/JCDL67857.2025.00024)

3. Kalir, H., German, F., Aboelnaga, Amr A., Banerjee, B., Eldardiry, H. & Ingram W. A. Retrieval-Augmented LLMs for ETD Subject Classification. In 2025, In 2025 IEEE International Conference on Big Data (BigData). [https://ai-collaboratory.net/wp-content/uploads/2025/11/S13211\\_3905.pdf](https://ai-collaboratory.net/wp-content/uploads/2025/11/S13211_3905.pdf)
4. Jadhav, A., Banerjee, B., & Goyne J. Using an Ensemble Approach for Layout Detection and Extraction from Historical Newspapers. In 2025 IEEE International Conference on Big Data (BigData). [https://ai-collaboratory.net/wp-content/uploads/2025/11/S13209\\_6075.pdf](https://ai-collaboratory.net/wp-content/uploads/2025/11/S13209_6075.pdf)
5. Miller, C., & Banerjee, B. Optical Character Recognition for Pre-Digital Historical Documents using Large Language Models. In 2025, The 29th International Conference on Machine Learning and Applications. <https://www.icmla-conference.org/icmla25/acceptedpapers.html/>. Accepted, proceedings not out yet.
6. Klair, H., German, F., Banerjee, B., & Ingram, W. A. (Accepted). Evaluating Human-LLM Alignment in ETD Subject Classification. In 2025, The 29th International Conference on Theory and Practice of Digital Libraries. [https://link.springer.com/chapter/10.1007/978-3-032-06136-2\\_6](https://link.springer.com/chapter/10.1007/978-3-032-06136-2_6)
7. Ingram, W. A., Banerjee, B., & Fox, E. A. (2025). When LLMs Disagree: Diagnosing Relevance Filtering Bias and Retrieval Divergence in SDG Search. LLM4Eval at SIGIR 2025 DOI: <https://doi.org/10.48550/arXiv.2507.02139>
8. Miller, C., Banerjee, B. (2025). Preserving and Understanding the Past through Large Language Models. Generative AI in Libraries Conference. URL: <https://shslibraryguides.org/genailibraries/schedule>
9. Banerjee, B., Goyne, J. (2025). AI-driven Solutions for Digital Library Collections. Generative AI in Libraries Conference. URL: <https://shslibraryguides.org/genailibraries/schedule>
10. Banerjee, B., Goyne, J., & Ingram, W. A. (2024, December). Making History Readable. In 2024 IEEE International Conference on Big Data (BigData) (pp. 8620-8622). IEEE. DOI: [10.1109/BigData62323.2024.10826028](https://doi.org/10.1109/BigData62323.2024.10826028)
11. Banerjee, B., Ingram, W. A., & Fox, E. A. (2024, December). Automating Chapter-Level Classification for Electronic Theses and Dissertations. In 2024 IEEE International Conference on Big Data (BigData) (pp. 2400-2409). IEEE. DOI: [10.1109/BigData62323.2024.10825418](https://doi.org/10.1109/BigData62323.2024.10825418)
12. Ingram, W. A., Banerjee, B., & Fox, E. A. (2024, December). Agentic AI for Improving Precision in Identifying Contributions to Sustainable Development Goals. In 2024 IEEE International Conference on Big Data (BigData) (pp. 8677-8679). IEEE. DOI: [10.1109/BigData62323.2024.10825072](https://doi.org/10.1109/BigData62323.2024.10825072)
13. Ingram, W. A., Wu, J., Kahu, S. Y., Manzoor, J. A., Banerjee, B., Ahuja, A., ... & Fox, E. A. (2024). Building datasets to support information extraction and structure parsing from electronic theses and dissertations. International Journal on Digital Libraries, 25(2), 175-196. DOI: <https://doi.org/10.1007/s00799-024-00395-4>
14. Park, S. H., Banerjee, B., Ingram, W. A., & Fox, E. A. (2023). Case Study of Analyzing the Variety of ETD Layouts. URL: <https://ir.inflibnet.ac.in/server/api/core/bitstreams/27afc90c-1a82-470e-8baa-5c1a3769426b/content>
15. Chekuri, S., Chandrasekar, P., Banerjee, B., Park, S. H., Masrourisaadat, N., Ahuja, A., ... & Fox, E. A. (2023, June). Integrated digital library system for long documents and their elements. In 2023 ACM/IEEE Joint Conference on Digital Libraries (JCDL) (pp. 13-24). IEEE. DOI: [10.1109/JCDL57899.2023.00012](https://doi.org/10.1109/JCDL57899.2023.00012)
16. Banerjee, B., Ingram, W. A., Wu, J., & Fox, E. A. (2022, December). Applications of data analysis on scholarly long documents. In 2022 IEEE International Conference on Big Data (Big Data) (pp. 2473-2481). IEEE. DOI: [10.1109/BigData55660.2022.10020935](https://doi.org/10.1109/BigData55660.2022.10020935)
17. Banerjee, B., Jude, P.M., Ingram, W.A., Luther K., Fox, E.A. (2022). Help Me Help You - A Mixed-Initiative Approach To Explore Book-length Documents. Workshop on Human-in-the-loop Data Curation at ACM CIKM 2022 <https://hilworkshops.github.io/hil-dc2022/>

18. Banerjee, B. (2022, June). Opening scholarly documents through text analytics. In Proceedings of the 22nd ACM/IEEE Joint Conference on Digital Libraries (pp. 1-2). DOI: <https://doi.org/10.1145/3529372.3530948>
19. Uddin, S., Banerjee, B., Wu, J., Ingram, W. A., & Fox, E. A. (2021, December). Building A large collection of multi-domain electronic theses and dissertations. In 2021 IEEE International Conference on Big Data (Big Data) (pp. 6043-6045). IEEE. DOI: [10.1109/BigData52589.2021.9672058](https://doi.org/10.1109/BigData52589.2021.9672058)
20. Banerjee, B., William A. Ingram, Jian Wu, and Edward A. Fox (2021), Applications of Mining ETDs. ETD 2021 Conference: Open scholarship in a post-pandemic world. DOI: <https://doi.org/10.26226/morressier.614c9b8c87a68d83cb5d59b2>
21. Aromando, J., Banerjee, B., Ingram, W. A., Jude, P., & Kahu, S. (2020). Classification and extraction of information from ETD documents. In Fox, E.A. course CS 6604: Digital Libraries, 2019. <http://hdl.handle.net/10919/96645>
22. Khawas, P., Banerjee, B., Kim, Y., Zhao, S., & Fan, Y. (2018). Big Data Text Summarization for Maryland Shooting Collection. In Fox, E.A. course CS 5984: Big Data Text Summarization 2018. <http://hdl.handle.net/10919/86407>
23. Ingram, W. A., Banerjee, B., & Fox, E. A. (2019). Summarizing ETDs with deep learning. Cadernos BAD, 1, 46-52. <https://opening-etds.github.io/files/2014-5500-1-PB.pdf>

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## RECENT GRANTS

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1. “Resilient Post-Industrial Landscapes along the James River: A Collaborative AI and Archival Approach” PI: Dr. Irene Curulli, School of Design, Co-PI: Dr. Bipasha Banerjee. University Libraries – Awarded \$10,000
2. “Human-Centered GeoAI for Age-Friendly Walkability at Warm Hearth Village” PI: Dr. Eunju Hwang, School of Design, Co-PIs: Dr. Bipasha Banerjee, Digital Library, Dr. Junghwan Kim, Geography. Co-I: Jennifer Goyne, Digital Library, Sara McCarter, WHV Community Partner, Dr. Jaehwan Kim, Real Estate, Kongju National University, South Korea - Awarded \$10,000

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## GRANTS SUBMITTED

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1. “Auditable AI Workflows for Evidence Synthesis Library Services”. PI: Bipasha Banerjee, Co-PI: C. Cozette Comer and William A. Ingram. Submitted as a two-year planning grant to Institute of Museum and Library Services for a total of \$200,000.
2. “Library-wide, Machine-Generated Metadata as First-Class Structural Components”. PI: Bipasha Banerjee, Co-PI: Jennifer L. Goyne, William A. Ingram. Submitted as a three-year applied research grant to the Institute of Museum and Library Services for a total of \$519,962.
3. “Unlocking the Past: Supporting Understanding and Insight of Historical Documents through AI-based Knowledge Extraction and Analysis” PI: Chreston A. Miller. Co-PI: Bipasha Banerjee, LaDale Winling. Submitted as a two-year research project to Schmidt Sciences Human and AI Virtual Institute program. \$423,544.21.

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## PATENTS

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Co-inventor on: US Patent Application 18/585,685; filed February 23, 2024; published 8/29/2024; Document ID US 20240289356 A1: Structured Documentation Access for Electronic Documents: Edward A. Fox, William A. Ingram, Aman Ahuja, Satvik Chekuri, and Bipasha Banerjee. <https://patents.google.com/patent/US20240289356A1>

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TEACHING EXPERIENCE

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**FACULTY MENTOR**

Mentoring graduate and undergraduate students on their research and curriculum work.

Virginia Tech, Blacksburg, VA, 2024 - Present

**GRADUATE STUDENT MENTOR**

Virginia Tech, Blacksburg, VA, 2019-2024

**GRADUATE TEACHING ASSISTANT**

Virginia Tech, Blacksburg, VA, JAN – MAY 2019

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PROFESSIONAL INVITED SERVICE

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- Invited reviewer for [Information and Culture](#), University of Texas Austin Press, June 2026
- Co-organized, developed and presented a pre-conference workshop titled “AI in Evidence Synthesis” at USAIN (US Ag Info Network) on May 2026.
- Invited Speaker for ETD 2026 Conference
- Co-chair, Conference Proceedings, IEEE/ACM Joint Conference on Digital Libraries, 2025, 2026.
- Invited reviewer for Springer Nature Scientometrics Journal Article, 2025.
- Invited as Program Committee Reviewer for IEEE/ACM Joint Conference on Digital Libraries 2022, 2023, 2024, 2025, 2026
- Co-organized a workshop titled “Hands-on digitizing texts with machine learning and AI”. A three-hour long workshop open to VT community on using machine learning and LLMs for optical character recognition

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PROFESSIONAL MEMBERSHIPS

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- Association for Computing Machinery (ACM) Member- #3768609
- Institute of Electrical and Electronics Engineers (IEEE) Member- #96669088

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PROFESSIONAL UNIVERSITY ACTIVITIES

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- Selected to serve as a member of library Continued Appointment Track Faculty Committee 2026-2028.
- Collaboration on a project geared toward using LLMs for University-wide analysis in partnership with University Libraries and the Office of Analytics & Institutional Effectiveness, Virginia Tech.
- Led breakout sessions at the workshop titled “Ensuring Scholarly Access to Government Archives and Records,” organized by Virginia Tech University Libraries in partnership with Virginia Tech Center for Humanities and the U.S. National Archives and Records Administration (funded by The Andrew W. Mellon Foundation)
- Digital Library Research Laboratory student representative for CS grad council at Virginia Tech (2020-2023).
- Association of Women in Computing Member, Virginia Tech Chapter.
- Volunteered at VT CS grad recruiting weekend and career fair.

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AWARDS

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- Nominated and inducted into Sigma Xi Scientific Honor Society - 2025
- Recognized as an inventor by Virginia Tech for “identifying and disclosing a technology for FY 23” in 2024.
- Nominated for best student paper for “Integrated Digital Library System for Long Documents and their Elements” at IEEE/ACM JCDL 2023.
- Awarded a scholarship to attend the Grace Hopper Conference by the CS department at Virginia Tech in 2021 and the Association of Women in Computing (AWC, Virginia Tech Chapter) in 2022.
- Awarded free attendance at the International Conference in Pattern Recognition (ICPR), virtual 2020.
- “Student of the Year” awarded by the Institute of Engineering and Management for academics as well as contribution to the college in 2014.